MODULE 2 · STATISTICS & PROBABILITY · TOPIC 2.8

Hypothesis Testing

Day 28 — is this difference real, or is it the √n noise we measured yesterday?



Boston University

1	4	3
Formula, built from yesterday's SE	Ways people misread a p-value	Habits that quietly break it

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WHY THIS SESSION

The maths is small. The misuse is enormous.

The whole formula

$$t = (\text{what we saw} - \text{what we expected}) / SE$$

That is it. The SE is yesterday's σ/√n. You already have every piece; today just adds a decision rule on top.

And the reason it needs two hours

This is the single most misused tool in applied statistics. Entire research literatures have been built on p-values that did not mean what their authors thought.

The formula takes ten minutes. Knowing when it lies to you takes the rest.

Four claims today, in order of how much they will bother you

1. A p-value is NOT the probability that your hypothesis is wrong
2. p = 0.049 and p = 0.051 differ by 0.017 in the test statistic
3. Two ordinary research habits triple your false positive rate
4. In some fields, most “significant” findings are false

WARM-UP • 5 MINUTES

Four questions from Day 27

1	**What is the standard error?**	→ σ/√n — how far a sample mean lands from the truth.
2	**σ or SE: which describes people?**	→ σ. SE describes how well you measured the average.
3	**Does the CLT make your data normal?**	→ No. It never did. It is about the MEAN.
4	**Name one way the CLT fails.**	→ Infinite variance, or dependent observations.

Everything today is built on that first answer. A hypothesis test asks one question: is the gap I observed large compared with the SE — the noise I already know how to compute?

SESSION PLAN

How today's two hours are structured

1. **The logic and the formula** — 15 min
   Assume nothing, measure surprise

2. **What a p-value is** — 20 min
   And the four things it is not

3. **Choosing the right test** — 20 min
   Welch, paired, and t vs z

4. **The two errors, and power** — 15 min
   Why “not significant” says little

5. **How it goes wrong** — 20 min
   Multiplicity, p-hacking, base rates

6. **Practice notebook** — 30 min
   You code, I walk around

Sections 2 and 5 are the ones that matter. Most courses teach section 1 and stop, which is how the misuse gets passed on.

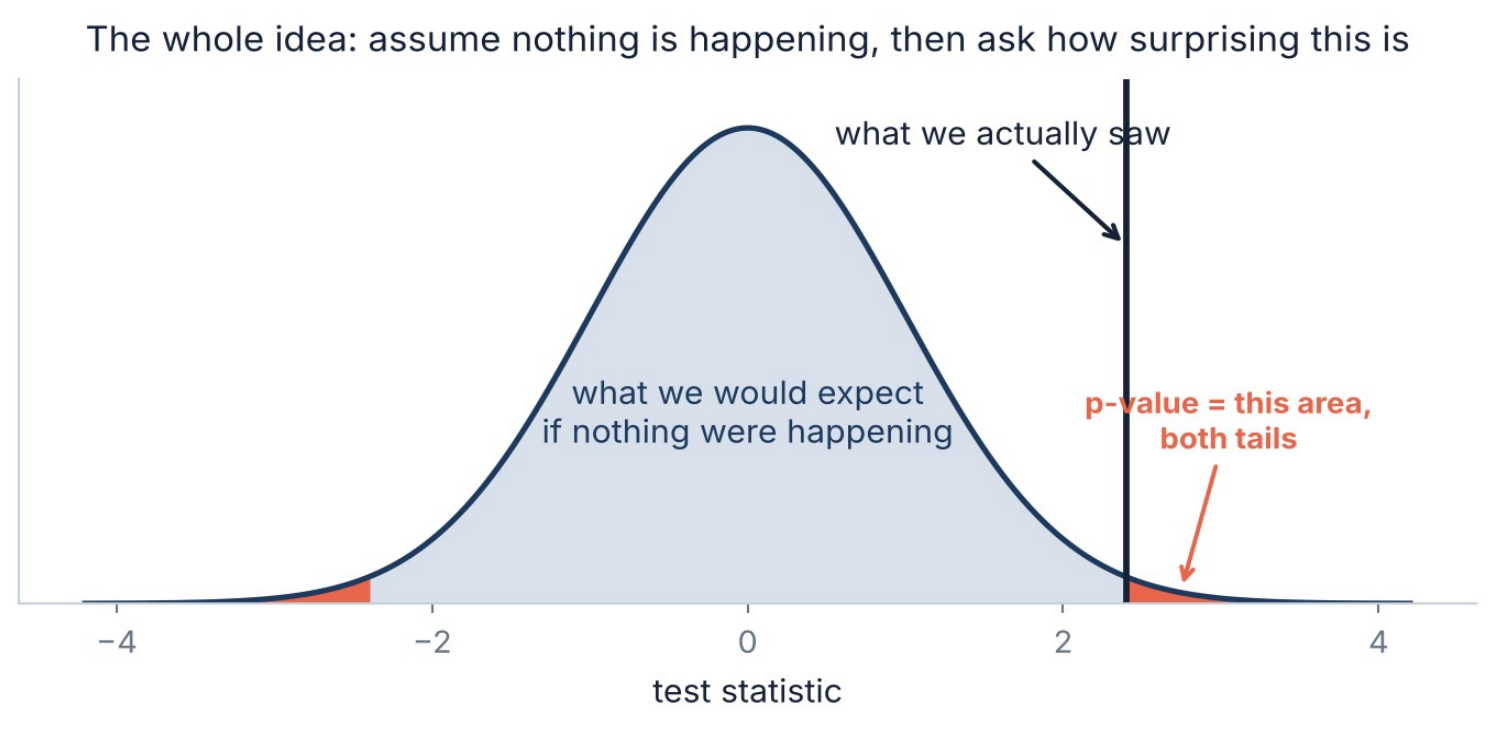
1

The Logic

Assume nothing, then measure surprise · 15 minutes

- The null hypothesis
- The formula
- A worked test

The whole idea, in one picture



Read it as a proof by contradiction, run on probabilities rather than logic: assume nothing is happening, work out what the data would look like under that assumption, and see whether what you actually got is a comfortable fit or an awkward one.

The formula is yesterday's standard error, rearranged

$$t = (\text{what we saw} - \text{what we would expect}) / SE$$

“how many standard errors away from boring is this?”

1	**State the null**	The boring version of the world: no difference, no effect, nothing happening.
2	**Compute the statistic**	The gap, divided by the SE you learned to compute yesterday.
3	**Find the p-value**	How often a world with no effect produces a gap at least this big.
4	**Decide — and report more**	The decision is the least interesting output. Report the effect and its interval too.

Worked through: did the training course help?

```
#  60 employees, productivity scored before and after a course.
#  Each person is measured twice, so the test is PAIRED.

change = after - before

#  --- 1. the null: the course does nothing, so mean change = 0
#  --- 2. the statistic
mean_d = change.mean()          #  +2.587
sd_d   = change.std(ddof=1)     #   5.978
se     = sd_d / np.sqrt(60)     #   0.772
t      = mean_d / se            #  +3.353

#  --- 3. the p-value
stats.ttest_rel(after, before)
#    t = 3.353,  df = 59,  p = 0.0014

#  --- 4. and the part that actually matters
stats.t.interval(.95, 59, mean_d, se)   # (+1.04, +4.13)
cohen_d = mean_d / sd_d                 #  0.43
```

Say it in words: “the course raised scores by about 2.6 points, and we are confident the true gain is between 1 and 4.”

PAIRED, not two-sample

same people measured twice

t = 3.35

3.35 standard errors from zero

p = 0.0014

rare, if the course did nothing

CI: +1.0 to +4.1 points

the useful output

d = 0.43

a modest, real-world effect

Report all three

p alone says almost nothing

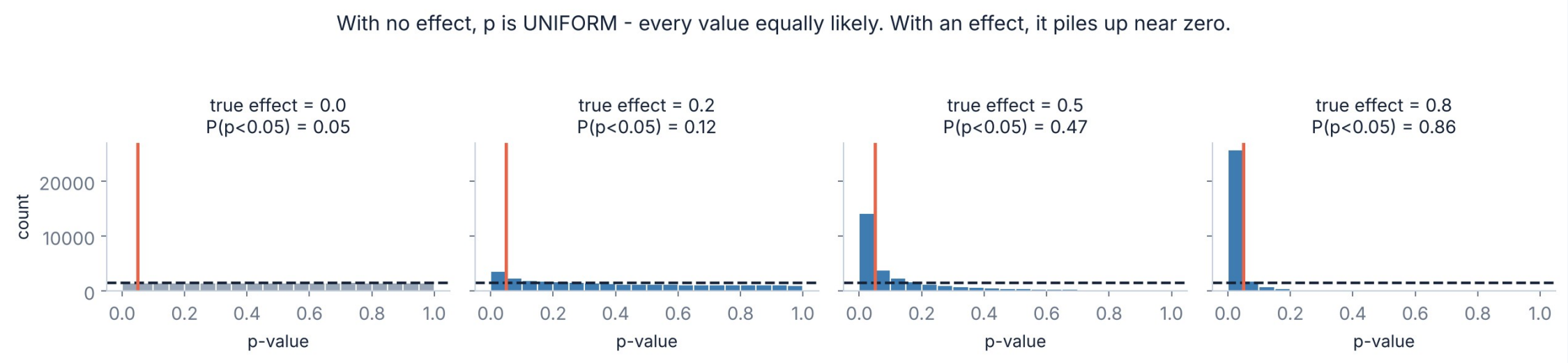
2

What a p-value Is

And the four things it is not · 20 minutes

- Uniform under the null
- The four misreadings
- The threshold is arbitrary

What a p-value does when nothing is happening



Left panel: no effect at all

The p-values are UNIFORM — 0.03 is exactly as likely as 0.97. That is Day 26's flat line, and it is the definition working: 5% of the time, a world with nothing in it hands you p < 0.05.

Right panels: a real effect

The distribution tips towards zero. A small p-value is evidence precisely because it is rare under the null — not because small numbers are meaningful in themselves.

⚠️ Four things a p-value is not

× "p = 0.03, so there is a 3% chance the null is true"

It is the other way round. p assumes the null IS true and asks how odd the data is. It cannot tell you the chance of the null.

× "p = 0.03, so there is a 97% chance the effect is real"

No. That depends on how plausible the effect was beforehand — section 5 puts a number on it.

× "p = 0.20, so there is no effect"

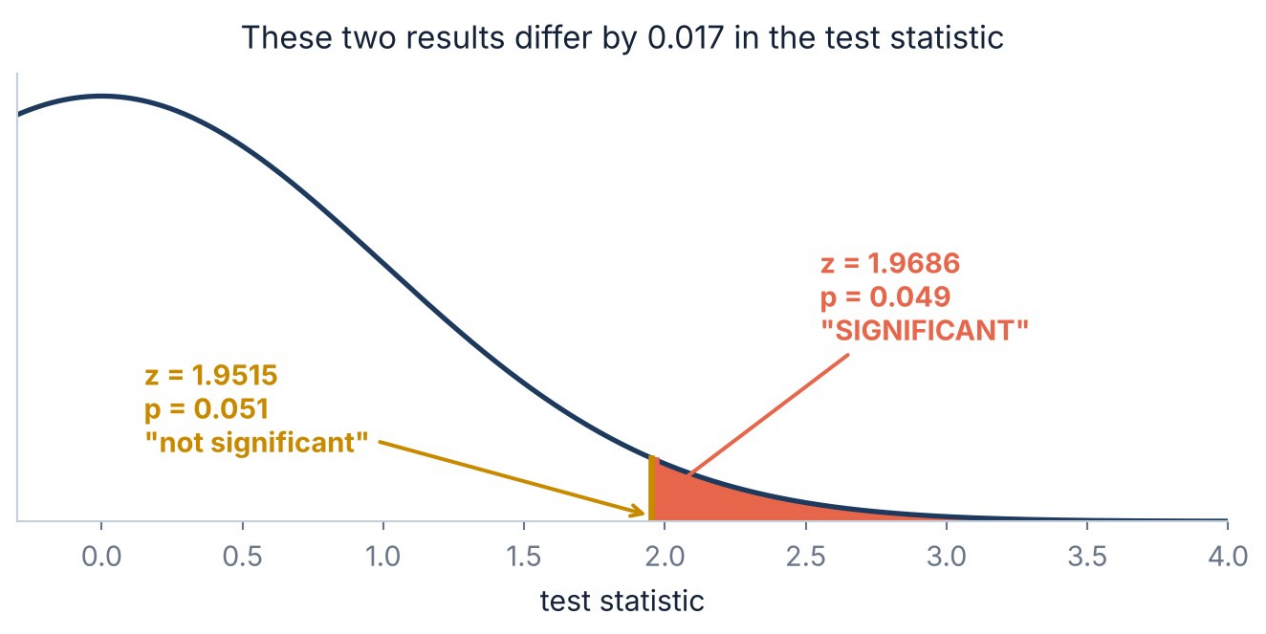
No. It means you did not find one, which may only mean your study was small. Absence of evidence.

× "p = 0.001, so the effect is large and important"

No. With enough data a meaningless difference gets a tiny p. Significance is about detectability, not size.

✓ **What it IS: if nothing were happening, how often would I see data at least this extreme? That sentence, and nothing more.**

⚠️ There is nothing magic about 0.05



The practical habit: report the actual p-value, not “p < 0.05”. A reader can then judge 0.049 and 0.004 differently — which they should, because they are different.

Two results, one career

p = 0.049 gets published and called a discovery. p = 0.051 gets a shrug and a file drawer.

The underlying test statistics are 1.9686 and 1.9515 — a difference of 0.017.

Where 0.05 came from

Ronald Fisher suggested it in 1925 as a convenient rule of thumb, roughly two standard errors. He explicitly intended it as a starting point, not a law.

Particle physics uses 0.0000003. The number should follow from the cost of being wrong.

3

Choosing the Test

Three choices that change the answer · 20 minutes

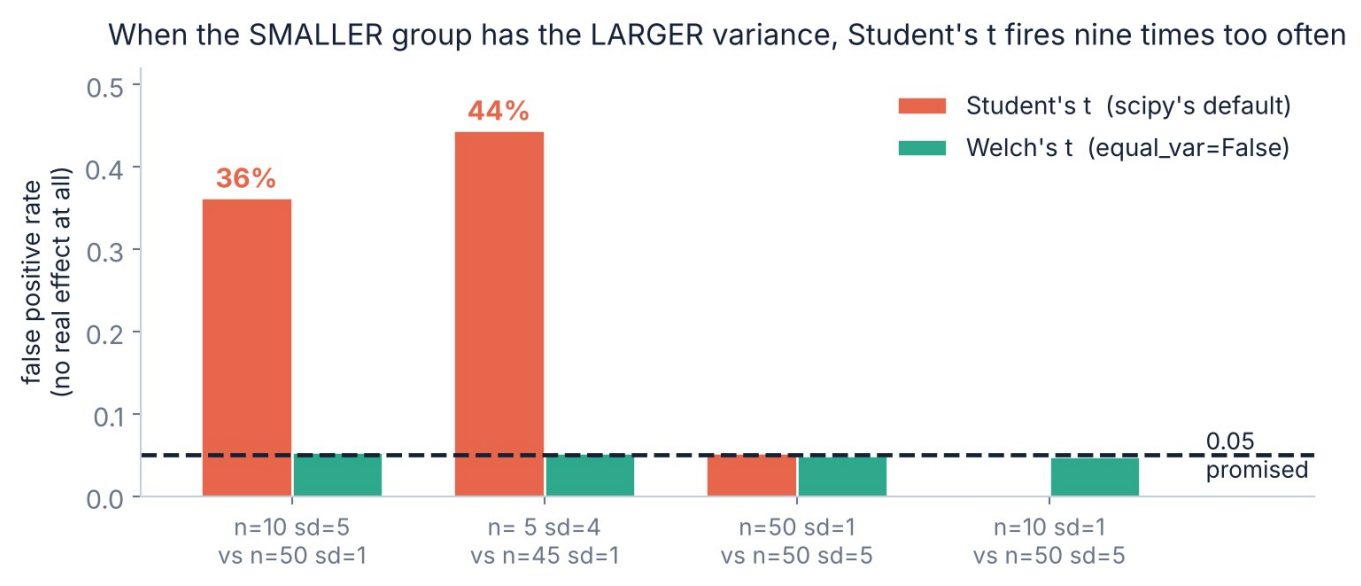
- Which test
- Welch vs Student
- Paired vs unpaired
- t vs z

Which test, in one table

Your question	The test	scipy
Is this mean different from a target?	one-sample t	`ttest_1samp(x, mu0)`
Are these two groups different?	Welch's t	`ttest_ind(a, b, equal_var=False)`
Same people, before and after?	paired t	`ttest_rel(after, before)`
Are these two proportions different?	two-proportion z	`proportions_ztest`
More than two groups?	ANOVA, then adjust	`f_oneway`
Counts in categories?	chi-square	`chi2_contingency`
Data badly non-normal, small n?	permutation test	`permutation_test`

The next three slides are the three choices in this table that most often get made wrong — and in each case the wrong choice runs without error and returns a confident number.

⚠️ scipy's default two-sample test is the wrong one



scipy's equal_var argument still defaults to True, so the dangerous version is the one you get by typing the obvious thing. Use Welch unless you have a specific reason not to — there is essentially no downside.

44% instead of 5%

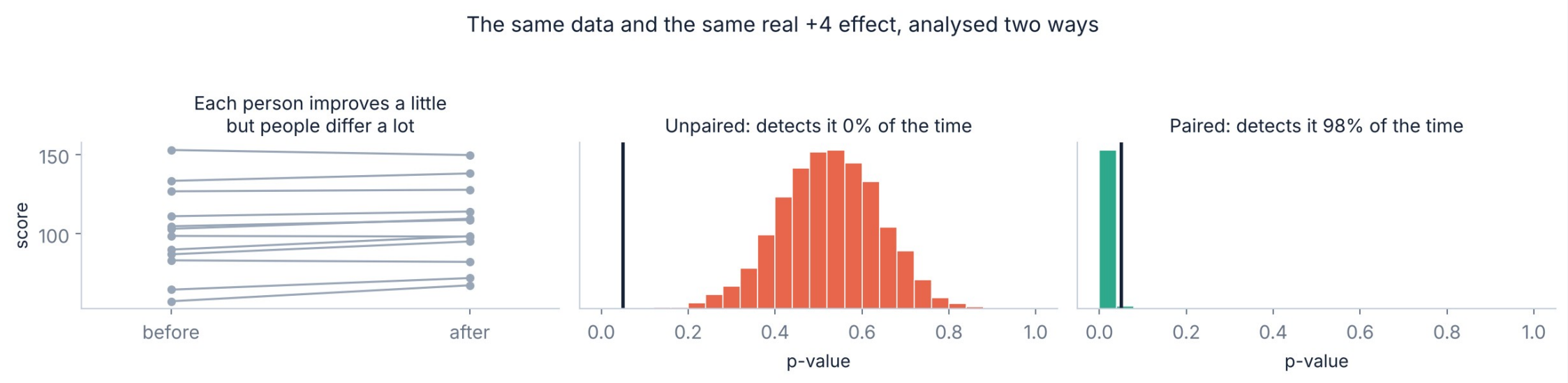
Student's t assumes both groups have the same variance. When the SMALLER group has the LARGER variance, that assumption fails badly and the test fires nine times too often — on data with no effect whatsoever.

The fix is one argument

```
stats.ttest_ind(
    a, b,
    equal_var=False)
```

Welch holds at 5% in every case above, and costs almost nothing when the variances do match.

Pairing can be the difference between 0% and 98%



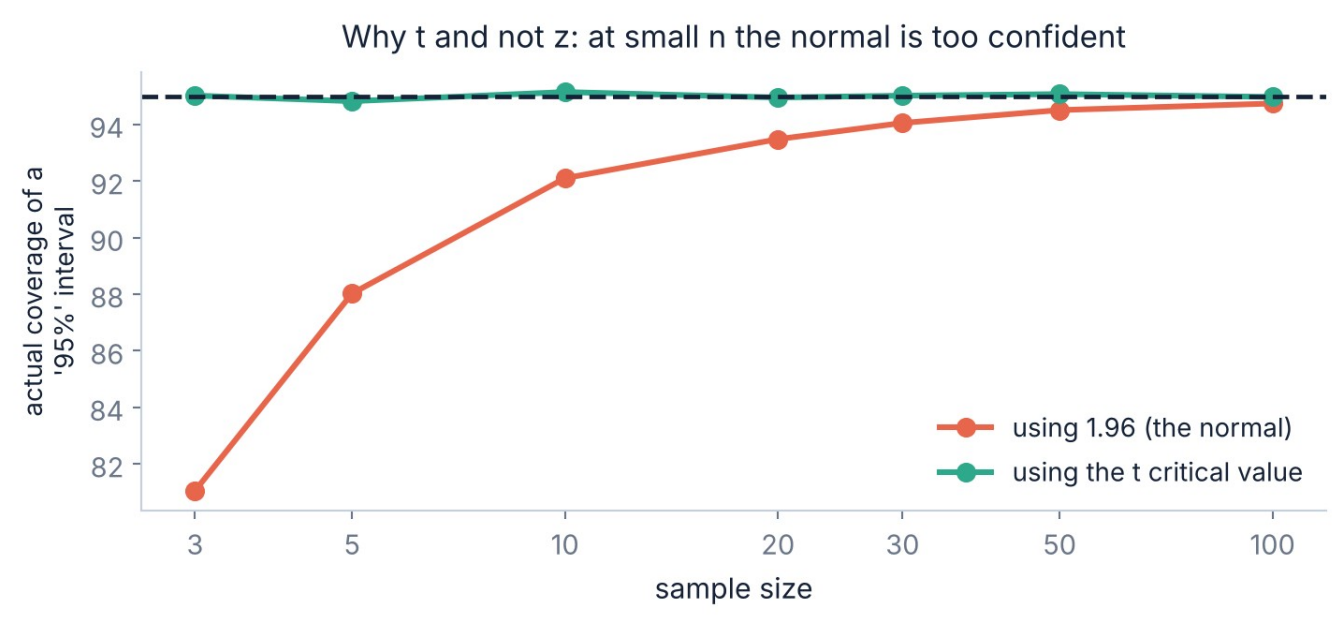
Why the unpaired test fails here

People differ by about 20 points; the effect is 4. An unpaired test has to see that 4-point signal through 20 points of between-person noise, and it cannot.

What pairing does

It subtracts each person from themselves. The between-person variation cancels completely, and all that is left is the change — which is exactly what you wanted to measure.

Why t and not z, and when it stops mattering



One extra source of error

When you use 1.96 you are pretending you know σ. You do not — you estimated it from the same small sample, and that estimate is itself uncertain.

The t distribution has heavier tails to pay for that.

How much it matters

n	using z	using t
3	81%	95%
10	92%	95%
30	94%	95%
100	95%	95%

Past about n = 100 the two agree to the third decimal, which is why nobody worries about it in large studies. Below n = 20 it is the difference between a 95% interval and a 92% one — and scipy's t functions handle it for you anyway.

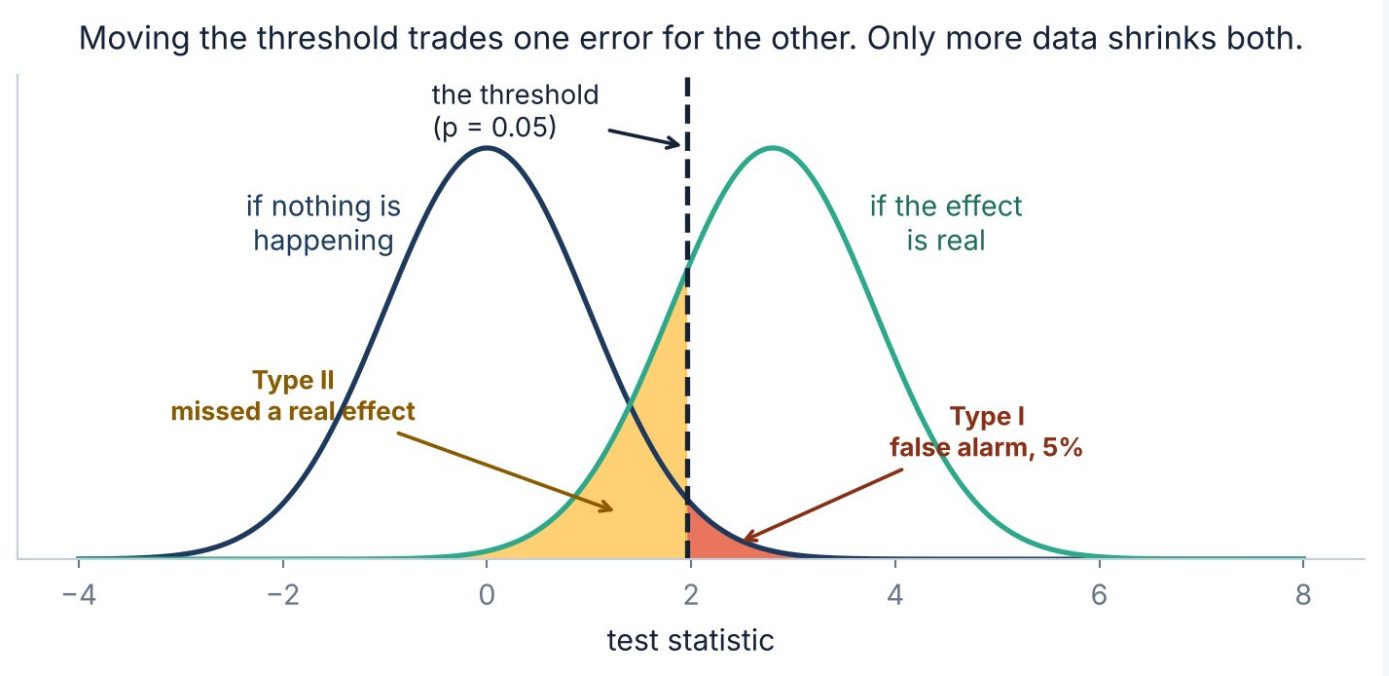
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The Two Errors

And why a null result says so little · 15 minutes

- Type I and Type II
- Power
- Absence of evidence

Two ways to be wrong, and they trade against each other



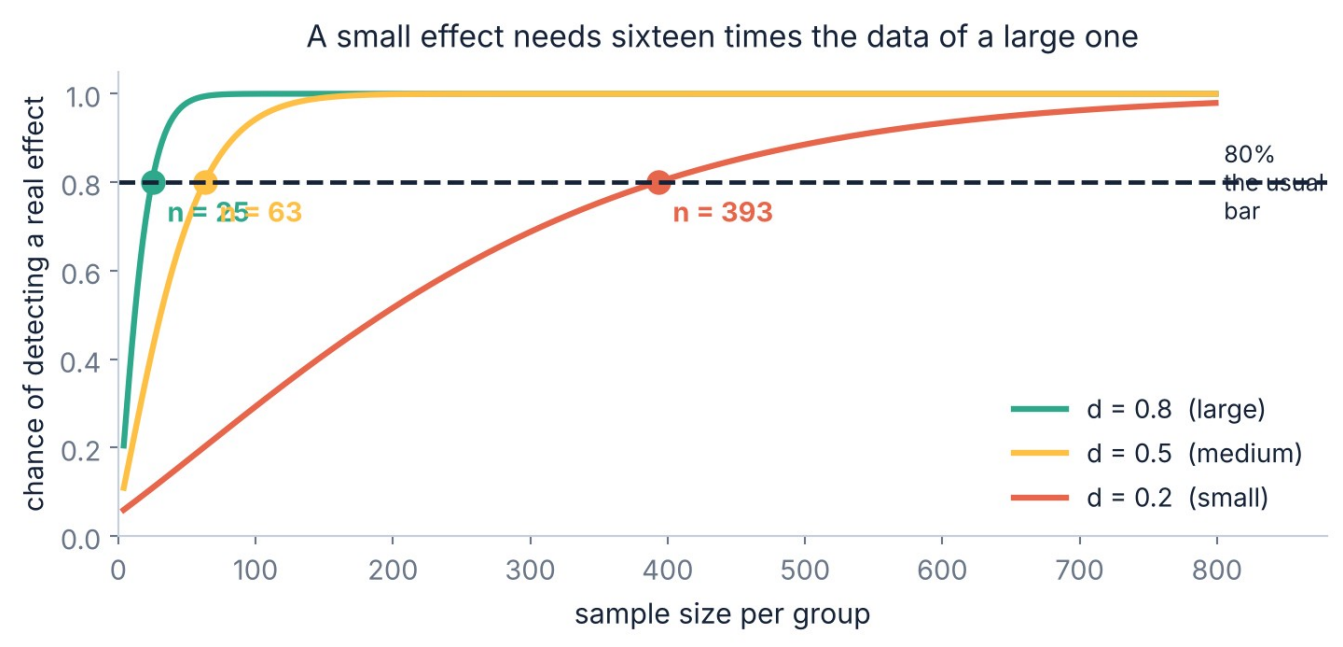
Type I — a false alarm

You declare an effect that is not there. You control this directly: it is α, the 5%.

Type II — a miss

A real effect goes undetected. You almost never control this, and almost nobody reports it.

Power: the chance of noticing an effect that is really there



Do this calculation BEFORE you collect data, not after. Afterwards it only tells you what you already know — that you did not find anything.

What the numbers mean

Cohen's d is the gap measured in standard deviations. 0.2 is small, 0.5 medium, 0.8 large.

For 80% power you need 25 per group at d = 0.8, 63 at d = 0.5, and 393 at d = 0.2.

Halve the effect, quadruple the n

25 → 63 → 393 as d goes 0.8 → 0.5 → 0.2. It is the same √n law as yesterday, seen from the other side: small effects are expensive to find.

⚠️ “Not significant” does not mean “no effect”

What a null result actually says

“We did not find an effect.” That is compatible with two completely different worlds:

- there is no effect, or
- there is one and our study was too small to see it.

The p-value alone cannot tell you which.

What to report instead

The confidence interval, which separates the two cases immediately:

- CI of −0.2 to +0.3 — any real effect is tiny. Informative.
- CI of −8 to +11 — you learned nothing at all. Also informative, and honest.

Same p-value. Completely different conclusions.

Why this matters beyond your own analysis

Journals have historically been reluctant to publish null results, so the studies that found nothing stay in the drawer while the ones that got lucky appear in print. The published record then looks far more certain than the evidence is. Reporting intervals rather than verdicts is the individual-level fix.

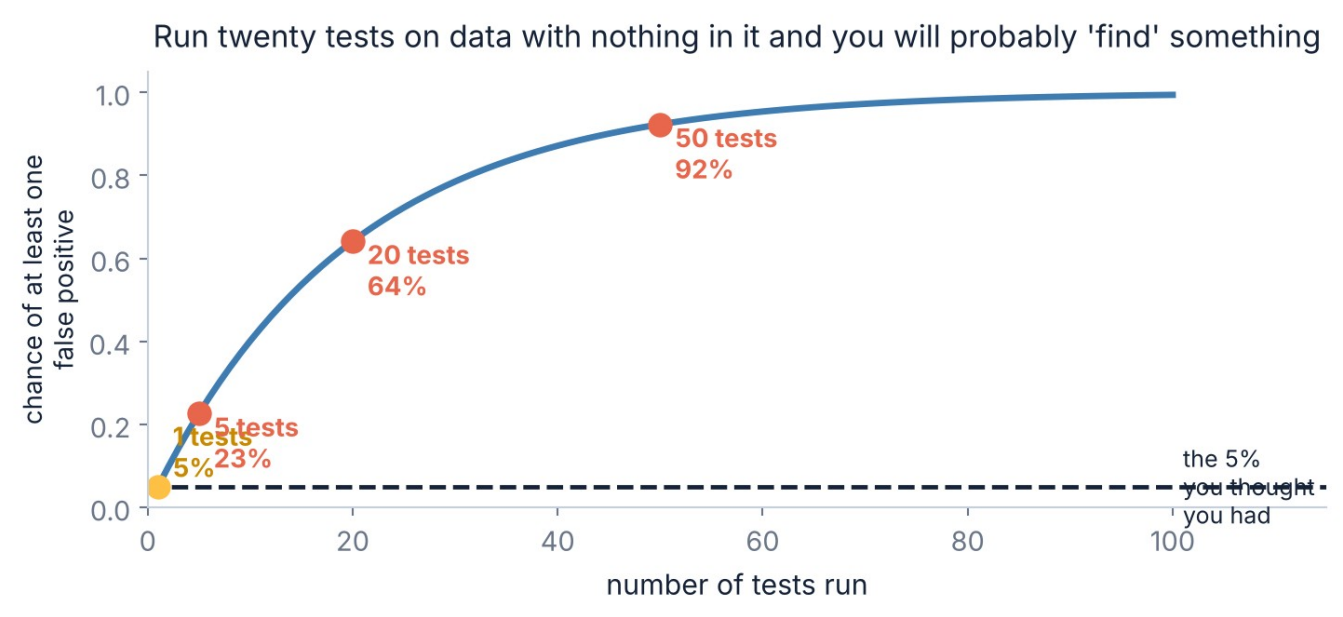
5

How It Goes Wrong

Three ways, none of which feel like cheating · 20 minutes

- Many comparisons
- p-hacking
- The base rate
- Significant but trivial

Run enough tests and something will always turn up



Where this bites in practice: testing ten metrics in an A/B test, comparing every pair of groups, or scanning a wide dataframe for "interesting" correlations. All three are routine.

Twenty tests, nothing real

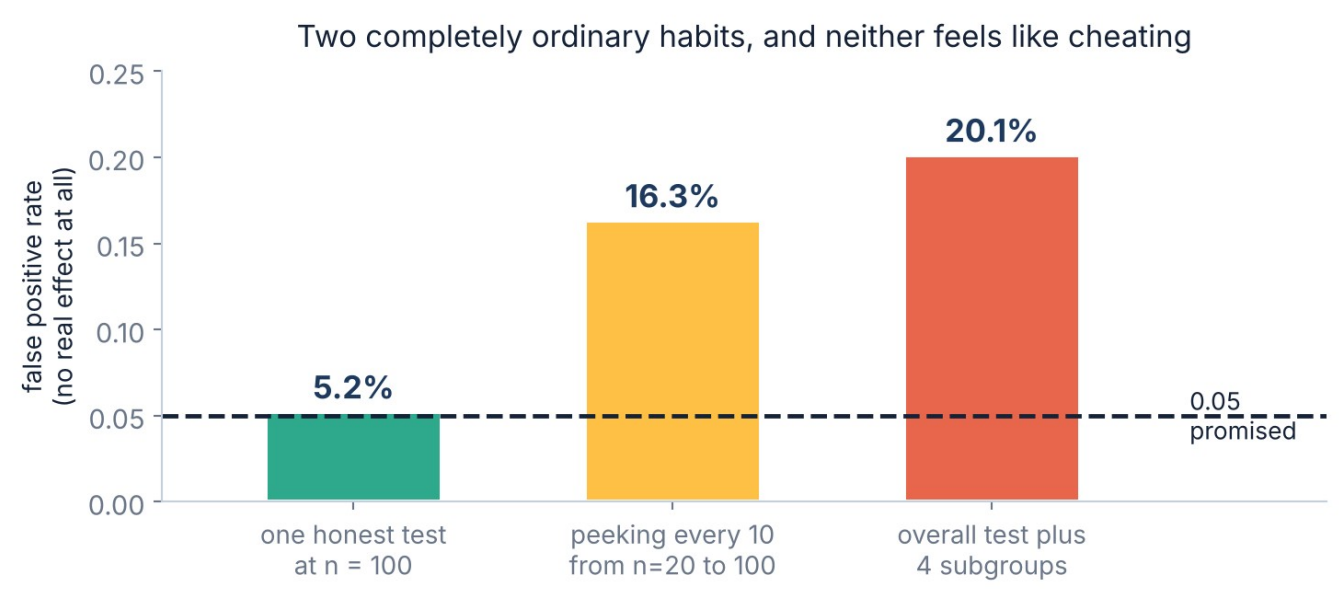
$1 - 0.95^{20} = 64\%$. On pure noise, you will find a "significant" result about two times in three, and on average exactly one.

This is not a subtle statistical issue. It is arithmetic.

The fixes

- Bonferroni: use 0.05/k. Simple, and strict.
- Benjamini–Hochberg: controls the false discovery rate. Less strict, usual choice for many tests.
- Best of all: decide which test matters before you look.

⚠️ Two habits that feel completely reasonable



Neither feels like cheating

"Let's check whether it's significant yet before spending more budget."

"It didn't work overall — but does it work for new users?"

Both are sensible-sounding sentences. Both triple or quadruple your false positive rate.

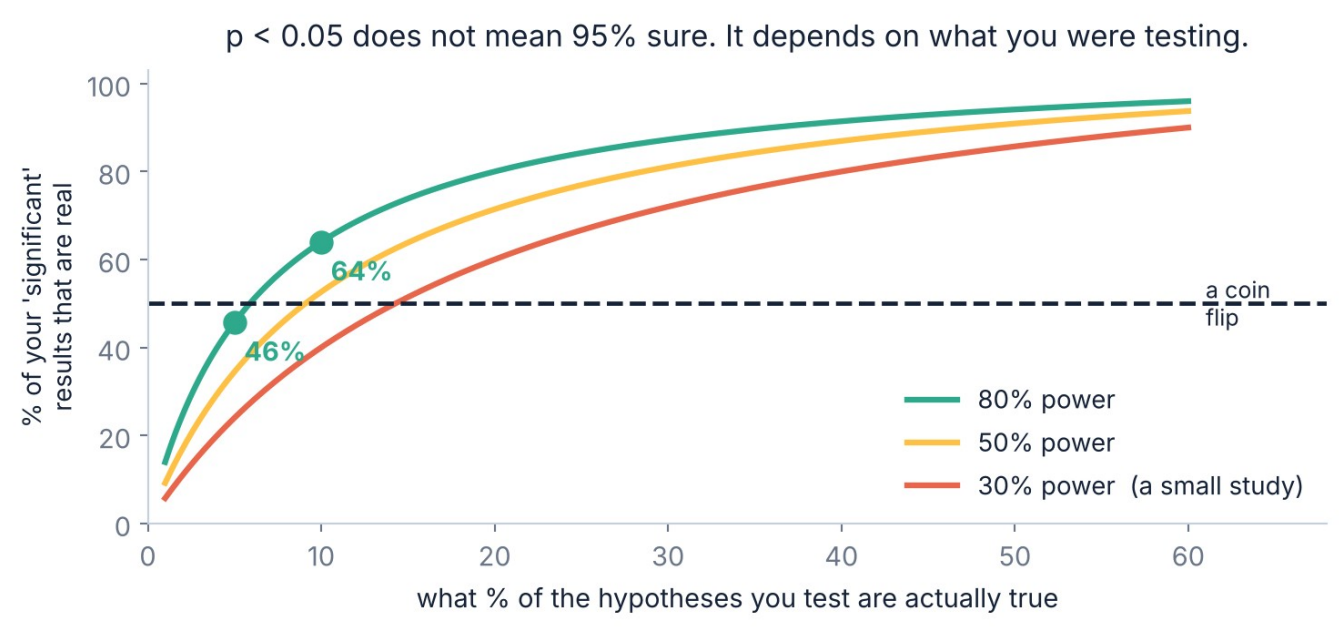
Why it happens

Every extra look is an extra chance for noise to cross the line. You are running many tests — you are just not counting them.

The fix: write down the analysis before you see the data, and count every test you run.

You will see this live in the practice notebook: a real +2.6 effect where the p-value goes 0.21 at n=10, then 0.041 at n=15, then back to 0.12 at n=25. Stopping at the good moment is the trap.

⚠️ p < 0.05 does not mean 95% sure



If only 5% of what you test is real — which is optimistic for exploratory work — then fewer than half your significant results are. The p-value cannot see this, because it never asked how plausible your hypothesis was.

The arithmetic

Test 1,000 hypotheses of which 100 are true, with 80% power:

- 80 true findings
- 45 false alarms (5% of 900)

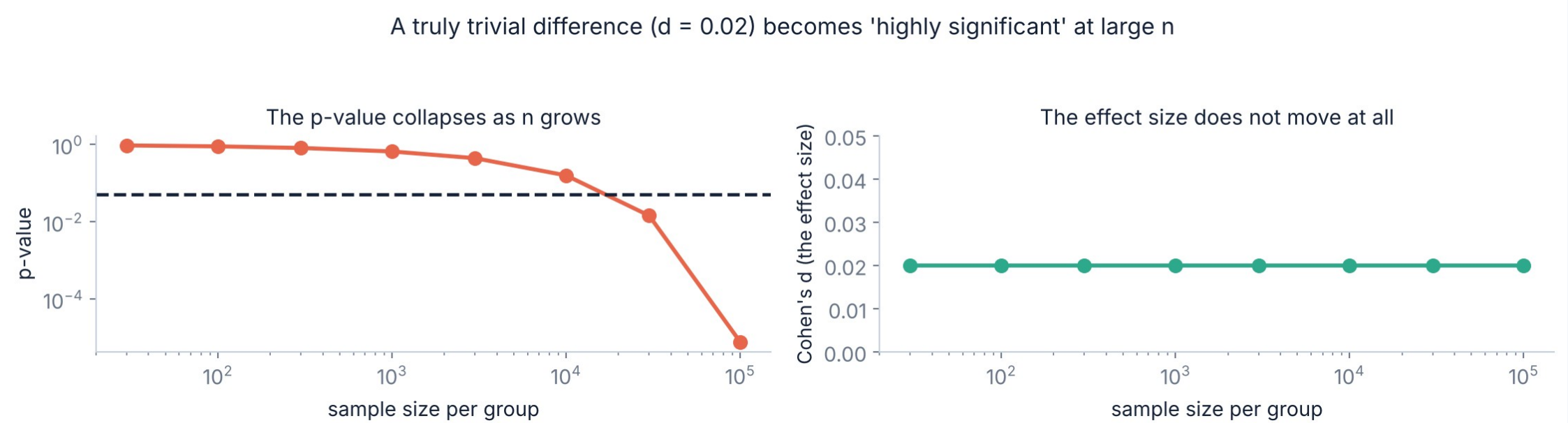
So 45 of your 125 “discoveries” — more than a third — are wrong.

What moves the number

- Testing plausible hypotheses, not fishing
- Higher power — bigger studies
- A stricter threshold where the cost of a false alarm is high

None of these is about the p-value itself.

⚠️ Significant means detectable, not important



Any difference becomes significant

A difference of d = 0.02 — utterly trivial — reaches p < 0.001 once n is large enough. No difference is exactly zero, so with enough data everything is “significant”.

The effect size does not care about n

Cohen's d is flat across the whole right-hand panel. It measures the thing you actually wanted to know, and it is the number to lead with.

What to report instead of a verdict

✕ The usual report

“The treatment group performed significantly better ($p < 0.05$).”

What a reader cannot tell from this: how big the effect was, how precisely it was measured, how many tests you ran, or whether the difference matters.

✓ A report worth reading

“The course raised scores by 2.6 points (95% CI 1.0 to 4.1; $d = 0.43$; paired t-test, $n = 60$, $p = 0.001$). This was the single pre-specified comparison.”

Every question above is answered, and the p-value is the least important thing in the sentence.

Five things to include, every time

• the effect size	• a confidence interval	• the test and n	• the exact p	• how many tests
in real units	the precision	reproducibility	not “< 0.05”	the honesty bit

The last one is the only one that requires any courage, and it is the one that makes the other four trustworthy.

RECAP

Debug corner — eight testing mistakes

You did	What goes wrong	Fix
read p as “chance the null is true”	it assumes the null, it cannot judge it	p is P(data | null), not P(null | data)
said “p > 0.05, so no effect”	you may just have had a small study	report the confidence interval
used ttest_ind with the default	equal_var=True fires 44% on some data	equal_var=False (Welch)
used ttest_ind on before/after	between-person noise swamps it	ttest_rel — the data is paired
tested 20 metrics, reported the best	64% chance of a false positive	Bonferroni or Benjamini-Hochberg
checked significance as data came in	5% becomes 16%	fix n in advance, or use a sequential design
reported p but not the effect size	a trivial difference looks like a finding	lead with the effect and its CI
said “p < 0.05”	0.049 and 0.004 are different	report the exact value

Rows 3 and 4 are one-word fixes that change the answer completely. Rows 5 and 6 are the ones that get published.

Putting it together — a test that reports honestly

```python
import numpy as np
from scipy import stats

def compare(a, b, n_tests=1, paired=False, label=""):
    """A two-group comparison that reports what matters."""
    a, b = np.asarray(a), np.asarray(b)

    if paired:
        d = b - a
        res = stats.ttest_rel(b, a)
        est, se, df = d.mean(), d.std(ddof=1)/np.sqrt(len(d)), len(d)-1
        cohen = d.mean() / d.std(ddof=1)
    else:
        res = stats.ttest_ind(a, b, equal_var=False)   # Welch
        est = b.mean() - a.mean()
        se  = np.sqrt(a.var(ddof=1)/len(a) + b.var(ddof=1)/len(b))
        df  = res.df
        cohen = est / np.sqrt((a.var(ddof=1) + b.var(ddof=1)) / 2)

    lo, hi = stats.t.interval(0.95, df, est, se)
    alpha  = 0.05 / n_tests                # Bonferroni

    print(f"{label} difference {est:+.3f}  95% CI [{lo:+.3f}, {hi:+.3f}]")
    print(f"  d = {cohen:+.3f}   p = {res.pvalue:.4f}   "
          f"threshold {alpha:.4f} ({n_tests} test(s))")
    print("  ->", "significant" if res.pvalue < alpha else "not significant")
```

Welch by default

the safe choice, not scipy's

n_tests is a required thought

you must say how many you ran

The CI comes first in the output

the effect leads, p follows

Effect size alongside

detectability vs importance

Exact p, never “< 0.05”

let the reader judge

It cannot stop you peeking

only your design can

Making n_tests a required argument is the whole trick: it forces you to answer the question nobody asks themselves.

OVER TO YOU

Practice session — 30 minutes

A **Predict the output** — 8 questions

p under the null, Welch, paired, power

B **Spot & fix the bug** — 4 questions

The default, the wrong test, a misread p

C **Write the code** — 5 tasks

Test, size, correct, and report

D **Challenge** — 2 tasks

p-hack on purpose, then audit a claim

Your notebook

`Day28_Practice_Questions.ipynb`

- Paired, or two groups?
- How many tests did I run?
- Report the effect, not the verdict
- Solutions shared after class

Part D asks you to p-hack a null dataset deliberately until you get a publishable result, and then to write the honest version of the same report. Doing it on purpose is the fastest way to recognise it being done to you.

TODAY IN TWELVE LINES

Recap

1. A test asks: is this gap big compared with yesterday's SE?
2. t = (what we saw − what we expected) / SE. That is the whole formula.
3. A p-value is P(data this extreme | nothing is happening). Nothing more.
4. Under the null, p is UNIFORM — which is why 5% of nothing looks like something.
5. ⚠️ p is NOT the chance the null is true, and NOT the chance you are right.
6. ⚠️ p = 0.049 vs 0.051 is a gap of 0.017 in the test statistic.
7. ⚠️ Use Welch (equal_var=False). scipy's default fires 44% on some data.
8. ⚠️ Paired data needs ttest_rel — it took one example from 0% to 98%.
9. Power is the chance of seeing a real effect. 80% needs n=25, 63 or 393 by size.
10. ⚠️ "Not significant" ≠ "no effect". Report the interval, which tells them apart.
11. ⚠️ 20 tests → 64% false positive. Peeking → 16%. Subgroups → 20%.
12. ⚠️ If 5% of your hypotheses are true, under half your "findings" are.

Homework

Find a published claim with a p-value and work out what is missing from the report · Take any two groups in your Day 21 project and compare them properly, reporting all five things · Work out the power your comparison had

Next class — Topic 2.9

Correlation. Measuring whether two variables move together — and the famous warning that goes with it, which turns out to be more interesting than the slogan suggests.